SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

COMPUTING ASSIGNMENT 1

– CO1

CO1 Coverage: Cellular network concepts; Frequency reuse and interference in cells; Access channels and mobile network capacity; Fundamentals of mobile computing

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Duration **:** 1 Week

Total Marks **:**50

Submission Date **:** 20/10/2026

Assignment Title: Design, Analyse and Optimise a Hexagonal Cellular Network for Spectrum Allocation, Interference Management and Scalable Coverage Planning in a University Township (Total Marks – 50)

Duration:1week

Scenario — Cellular Coverage Planning for a University Township

An operator has been licensed to provide cellular service to a 12 km² university township containing academic blocks, hostels, residential colonies and a main arterial road. The operator holds a fixed allocation equivalent to 336 duplex voice channels (after control-channel overhead) and must plan a hexagonal-cell network. Subscriber demand will grow as the township expands.

You are the radio-planning team. You must investigate how spectrum can be shared across cells, how interference limits that sharing, and how the design must change as the subscriber base grows.

Planning data and constraints

- Subscribers: 1,000 → 10,000 → 20,000 → 100,000+ (planning horizon).
- Average busy-hour traffic per subscriber: 0.025 Erlang. Target grade of service: ≤ 2% call blocking.

- Minimum acceptable SIR: choose and justify a threshold between 12 dB (digital-style) and 18 dB (analogue-style).
- Path-loss exponent n: 2.5 (open campus) to 4 (dense built-up area); justify the values you adopt.
- Spectrum cannot be increased; only the network layout and radio parameters may change.
- Any planning effort (surveys, traffic estimation, number of sites) must be included in your cost analysis.

Design Challenge (Day 4):

A convocation / sports event will bring 25,000 attendees (0.05 Erlang each) into a 1 km^2 stadium zone for four hours. Re-evaluate your original design and decide whether to retain, modify or redesign it.

TASK 1 – Design Alternative Frequency-Planning Strategies 5 Marks

Guiding Enquiry

How can a fixed block of spectrum be shared across many cells so that every subscriber is served without intolerable co-channel interference — and in how many fundamentally different ways can this be done?

Investigate and interpret:

- Propose at least **three meaningfully different strategies** (for example fixed cluster reuse, sectorised reuse, dynamic/borrowed channel assignment, hierarchical macro–micro layering). The same strategy with different cluster sizes (N = 3 vs N = 7) does not count as two strategies.
- At least one strategy must be expressed as an **iterative procedure** (e.g., searching valid cluster sizes $N = i^2 + ij + j^2$ and assigning channel sets across the grid) and at least one as a **recursive/hierarchical procedure** (e.g., repeated cell splitting until demand is met).
- analyse the basic idea, the assumptions, and the way the **access channels** are shared (FDMA/TDMA/CDMA/OFDMA concept and control vs traffic channels).
- Provide pseudocode or a flowchart for each, name the design principle used (spatial reuse, trunking, spatial filtering), and justify why each deserves investigation.
- State the preprocessing required (traffic map, propagation survey, hex-grid layout) and who bears that cost.

EVIDENCE EXPECTED

Three distinct strategies with pseudocode/flowcharts, identified design principle, preprocessing requirements and justification for each.

ANSWER:

TASK 1 – DESIGN ALTERNATIVE FREQUENCY-PLANNING STRATEGIES (5 MARKS)

AIM:

To design three different frequency-planning strategies for sharing limited spectrum among cellular cells while reducing co-channel interference.

1. FIXED CLUSTER FREQUENCY REUSE

DESIGN PRINCIPLE: SPATIAL REUSE

- Divide the township into hexagonal cells.
- Group seven cells into one cluster (N = 7).
- Total available channels = 336.
- Assign different frequency channels to each cell.
- Each cell receives: S=336/7=48
- Repeat the same frequency pattern across the network.

PSEUDOCODE:

START

Set total channels = 336

Set cluster size N = 7

Calculate channels per cell

Create hexagonal cells

Assign different channels to cells

Repeat cluster pattern

Check interference

STOP

2. SECTORISED FREQUENCY REUSE

DESIGN PRINCIPLE: SPATIAL FILTERING

- Divide each hexagonal cell into three 120° sectors.
- Install directional antennas at the base station.
- Distribute channels among the sectors.
- Each sector receives approximately: S=48/3=16
- Directional antennas reduce co-channel interference.

PSEUDOCODE:

START

Create hexagonal cells

Divide each cell into 3 sectors

Install directional antennas

Assign frequency channels

Measure SIR

Adjust antenna direction

STOP

3. DYNAMIC CHANNEL ASSIGNMENT

DESIGN PRINCIPLE: TRUNKING EFFICIENCY

- Maintain a common pool of 336 channels.
- Assign channels based on traffic demand.
- Check channel availability before assignment.

- Borrow unused channels from neighbouring cells.
- Block calls when no suitable channel is available.

PSEUDOCODE:

START

Create common channel pool

Receive incoming call

Check free channels

Check interference

IF channel is available

    Assign channel

ELSE

    Borrow channel from neighbour

    IF unavailable

        Block call

STOP

4. PYTHON PROGRAM

SOFTWARE USED: Python 3, VS Code.

```
channels = 336

N = 7

print("Fixed Cluster Reuse")

ch = channels // N

for i in range(1, N + 1):
```

```
    print("Cell", i, "Channels:", ch)
print("\n Sectorised Reuse")
for i in range(1, 4):
    print("Sector", i, "Channels:", ch // 3)
print("\n Dynamic Channel Assignment")
available = list(range(1, channels + 1))
for cell in range(1, 6):
    if available:
        print("Cell", cell, "Channel:", available.pop(0))
    else:
        print("Call Blocked")
print("\nPlanning Completed")
```

OUTPUT:

```
Fixed Cluster Reuse
Cell 1 Channels: 48
Cell 2 Channels: 48
Cell 3 Channels: 48
Cell 4 Channels: 48
Cell 5 Channels: 48
Cell 6 Channels: 48
Cell 7 Channels: 48
Sectorised Reuse
```

Sector 1 Channels: 16

Sector 2 Channels: 16

Sector 3 Channels: 16

Dynamic Channel Assignment

Cell 1 Channel: 1

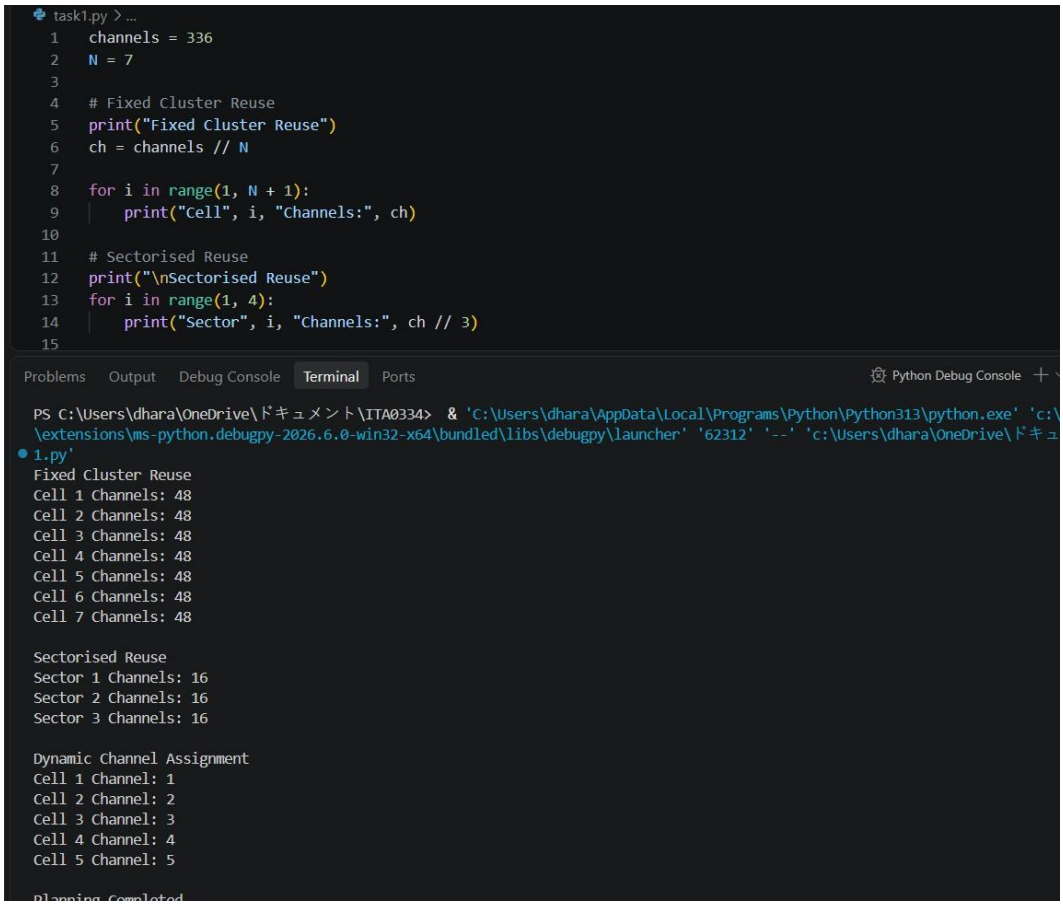
Cell 2 Channel: 2

Cell 3 Channel: 3

Cell 4 Channel: 4

Cell 5 Channel: 5

Planning Completed



5. COMPARISON OF STRATEGIES

Strategy	Principle	Advantage
Fixed cluster reuse	Spatial reuse	Simple planning
Sectorised reuse	Spatial filtering	Reduces interference
Dynamic assignment	Trunking efficiency	Better channel utilisation

6. PREPROCESSING REQUIREMENTS

1. Collect subscriber traffic information.
2. Conduct radio propagation surveys.
3. Prepare a hexagonal cell layout.
4. Identify buildings and interference areas.
5. Estimate the number of base stations required.

COST: The network operator bears the survey, traffic estimation, planning and installation costs.

RESULT:

Three frequency-planning strategies were designed and demonstrated using Python. Fixed cluster reuse, sectorised reuse and dynamic channel assignment help utilise limited spectrum efficiently, reduce interference and support increasing subscriber demand.

TASK 2 – Identify the Governing Design Parameter **2 Marks**

Guiding Enquiry

For each strategy, which single quantity most strongly controls capacity and interference, and why should the analysis be built around it?

Investigate and interpret:

- Identify the dominant parameter(s) for each strategy (e.g., cluster size N, reuse ratio Q = D/R, number of co-channel interferers, cell radius R, channels per cell).
- analyse where it appears in the strategy and why it is appropriate for efficiency analysis.
- interpret how this quantity must change as subscribers grow from 1,000 to 100,000+.

Evidence Expected

Correct identification and justification of the governing parameter for each strategy, linked to its behaviour.

ANSWER:

1. INTRODUCTION

The governing design parameter is the most important quantity that determines the capacity and interference of a cellular network. Each frequency-planning strategy has a different governing parameter. These parameters help in deciding how frequencies are allocated, how interference is controlled and how network capacity is increased when the number of subscribers grows.

2. GOVERNING PARAMETERS FOR EACH STRATEGY

1. FIXED CLUSTER REUSE – CLUSTER SIZE (N)

- Cluster size (N) represents the number of cells in a frequency reuse group.
- It determines how many channels are allocated to each cell.
- A smaller cluster size increases the number of channels per cell but also increases interference.
- A larger cluster size reduces interference but decreases the number of channels available per cell.

Formula:

Channels per cell ={S}/{N}

Where:

- S = Total available channels (336)
- N = Cluster size.

2. SECTORISED REUSE – NUMBER OF CO-CHANNEL INTERFERERS

- Sectoring divides a cell into smaller sectors using directional antennas.
- It reduces the number of interfering signals received by a mobile user.
- Fewer interferers improve the Signal-to-Interference Ratio (SIR).
- Therefore, sectoring improves network performance and supports better frequency reuse.

3. DYNAMIC CHANNEL ASSIGNMENT – AVAILABLE CHANNELS

- Channels are assigned dynamically based on current traffic demand.
- The system selects an available channel whenever a subscriber makes a call.
- It improves channel utilisation and reduces call blocking.
- The available channel pool is the main parameter controlling the capacity.

4. **HIERARCHICAL MACRO–MICRO CELL – CELL RADIUS (R)**

- Cell radius determines the geographical coverage of each base station.
- Reducing cell radius creates more cells in the same geographical area.
- More cells allow greater frequency reuse and increase network capacity.
- However, smaller cells increase handoff frequency and infrastructure cost.

3. EFFECT OF SUBSCRIBER GROWTH

The network parameters must be adjusted according to the increase in subscribers.

Subscribers	Required Network Changes
1,000	Small number of cells with basic frequency reuse is sufficient.
10,000	Increase channel utilisation and optimise frequency allocation.
20,000	Introduce sectoring and smaller cells to improve capacity.
100,000+	Use microcells, dynamic channel assignment and repeated cell splitting.

4. IMPORTANCE OF GOVERNING PARAMETERS

The governing parameters are important for the following reasons:

1. They help maximise the use of the fixed 336 available channels.
2. They control co-channel interference between neighbouring cells.
3. They improve network capacity as subscriber demand increases.
4. They help maintain the required Signal-to-Interference Ratio (SIR).
5. They reduce call blocking and improve the quality of mobile communication.

5. CONCLUSION

The main governing parameters are cluster size (N), number of co-channel interferers, available channels and cell radius (R). These parameters directly influence network capacity, frequency reuse and interference. By adjusting them according to subscriber demand, the cellular network can provide efficient coverage and reliable communication from 1,000 to 100,000+ subscribers.

TASK 3 – Derive the Capacity and Interference Functions Mathematically 6 Marks

Guiding Enquiry

Starting from hexagonal geometry, how do coverage, capacity and interference depend on the design choices you made?

Investigate and interpret:

- Derive the cluster-size relation, reuse distance $D = R\sqrt{(3N)}$ and reuse ratio Q, hexagonal cell area, and the number of cells needed to cover 12 km².
- Derive channels per cell (S/N), traffic per cell at 2% blocking (Erlang-B reasoning), and total system capacity as a function of N and R.
- Derive the co-channel SIR expression for each strategy (first-tier interferers; effect of sectoring on the number of interferers).
- Combine **planning cost** (survey, number of sites) with **operating capacity** to obtain an overall cost–capacity relation for each strategy.
- Show every step; stating a final formula without derivation will not earn full credit.

Evidence Expected

Clear derivations of capacity, number of cells and SIR for each strategy, including planning cost and overall cost–capacity relation.

ANSWER

TASK 3 – DERIVE CAPACITY AND INTERFERENCE FUNCTIONS MATHEMATICALLY

1. INTRODUCTION

The capacity and interference of a cellular network depend on frequency reuse, cell radius, cluster size, available channels and traffic demand. Mathematical models are used to calculate the number of cells, channels per cell, call capacity and signal-to-interference ratio (SIR).

Given Data:

- Total available duplex channels (S) = 336
- Total township area (A) = 12 km²
- Busy-hour traffic per subscriber = 0.025 Erlang
- Maximum call blocking probability = 2%
- Path-loss exponent (n) = 2.5 to 4
- Assumed cell radius (R) = 1 km
- Cluster size (N) = 7

2. FREQUENCY REUSE AND CLUSTER SIZE

The cluster size is calculated using:

$N = i^2 + ij + j^2$

Where:

- i and j are shift parameters.
- N is the number of cells in one frequency reuse cluster.

For example, take i = 2 and j = 1.

$N = 2^2 + (2 \times 1) + 1^2$

$N = 7$

Therefore, the cluster contains 7 cells.

3. FREQUENCY REUSE DISTANCE

The distance between two co-channel cells is calculated using:

$D = R \times \sqrt{(3N)}$

Substituting R = 1 km and N = 7:

$D = 1 \times \sqrt{(3 \times 7)}$

$D = 4.58$ km

Therefore, the frequency reuse distance is approximately 4.58 km.

4. FREQUENCY REUSE RATIO

The frequency reuse ratio is calculated using:

$Q = D / R$

$Q = 4.58$

A higher reuse distance helps reduce co-channel interference but may reduce the number of channels available per cell.

5. HEXAGONAL CELL AREA

The area of a regular hexagonal cell is:

$Ac = (3 \times \sqrt{3} / 2) \times R^2$

Substituting R = 1 km:

$Ac = (3 \times 1.732 / 2) \times 1^2$

$Ac = 2.598 \text{ km}^2$

Therefore, the area of one hexagonal cell is approximately 2.598 km^2.

6. NUMBER OF CELLS REQUIRED

The number of cells required to cover the township is:

M = Total area / Area of one cell

M = 12 / 2.598

M = 4.62

Rounding up to the next whole number:

M = 5 cells

Therefore, approximately 5 cells are required to cover the 12 km^2 township area.

Note: This is an ideal area-based estimate. Actual hexagonal placement may require additional cells because of boundary coverage and overlap.

7. CHANNELS PER CELL

The number of channels allocated to each cell is:

Channels per cell = Total available channels / Cluster size

C = S / N

C = 336 / 7

C = 48 channels per cell

Therefore, each cell receives 48 duplex voice channels under the fixed frequency reuse strategy.

8. ERLANG-B TRAFFIC CAPACITY

The Erlang-B formula is used to calculate the maximum traffic that can be supported for a given number of channels and blocking probability.

Erlang-B formula:

B(A,C) = (A^C / C!) / [Σ (A^k / k!), from k = 0 to C]

Where:

- B = Call blocking probability
- A = Offered traffic in Erlangs
- C = Number of channels
- k = Summation index

Given:

C = 48 channels

B = 0.02 (2% blocking probability)

Using the Erlang-B calculation:

A ≈ 38.39 Erlangs per cell

Therefore, one cell can support approximately 38.39 Erlangs of offered traffic at a 2% blocking probability.

9. TOTAL NETWORK CAPACITY

The total network traffic capacity is:

Total capacity = Number of cells × Traffic capacity per cell

A = 5 × 38.39

A = 191.95 Erlangs

The number of subscribers supported is:

Subscribers = Total traffic capacity / Traffic per subscriber

Subscribers = 191.95 / 0.025

Subscribers = 7,678

Therefore, the estimated network capacity is approximately 7,678 subscribers under the assumed parameters.

10. SIGNAL-TO-INTERFERENCE RATIO (SIR)

SIR measures the strength of the desired signal compared with co-channel interference.

For a hexagonal cellular network with six first-tier co-channel interferers:

SIR = (D / R)^n / 6

Since D / R = √(3N):

SIR = (3N)^(n/2) / 6

For N = 7 and n = 4:

SIR = (3 × 7)² / 6

SIR = 73.5

Converting into decibels:

SIR (dB) = 10 × log10(73.5)

SIR = 18.66 dB

Therefore, the calculated SIR is approximately 18.7 dB, which satisfies the 18 dB target under the assumed ideal conditions.

11. SIR WITH SECTORING

Sectoring divides a cell into smaller coverage regions using directional antennas. It reduces the number of dominant co-channel interferers.

Assuming 120-degree sectoring reduces the dominant interferers from six to two:

SIR = (3N)^(n/2) / 2

Substituting N = 7 and n = 4:

SIR = 21^2 / 2

SIR (dB) = 10 × log10(220.5)

SIR = 23.43 dB

Therefore, sectoring improves the estimated SIR from 18.66 dB to 23.43 dB.

Note: This is an idealized calculation assuming two dominant interferers.

12. CAPACITY AND INTERFERENCE FUNCTIONS FOR DIFFERENT STRATEGIES

A. FIXED FREQUENCY REUSE

Channels per cell = S / N

SIR = (3N)^(n/2) / 6

A larger cluster size improves SIR but reduces the number of channels available per cell.

B. SECTORISED FREQUENCY REUSE

SIR = (3N)^(n/2) / I

Where I is the number of dominant co-channel interferers.

Sectoring reduces interference and improves signal quality.

C. DYNAMIC CHANNEL ASSIGNMENT

Channels are allocated according to current traffic demand and channel availability.

Channels per cell = Number of channels available at the time of allocation.

This method improves channel utilisation and reduces congestion.

D. MACRO–MICRO CELL HIERARCHY

The number of cells required is:

M = A / [(3 × √3 / 2) × R^2]

Reducing the cell radius increases the number of cells and improves spatial frequency reuse. However, it also increases infrastructure and handoff requirements.

13. NETWORK PLANNING COST

The total network cost can be calculated using:

C_T = C_s + (M × C_b) + C_o

Where:

- C_T = Total network cost
- C_s = Survey and planning cost
- M = Number of cells
- C_b = Cost per base station
- C_o = Operating cost

COST PER ERLANG

Cost per Erlang = C_T / A_T

Where A_T is the total network traffic capacity.

A network with more cells provides higher capacity but requires additional installation, operation and maintenance costs.

14. CONCLUSION

The mathematical analysis shows that cellular network capacity depends on cell radius, available channels, cluster size and traffic demand. Frequency reuse improves spectrum efficiency, while sectoring reduces co-channel interference. The Erlang-B model helps maintain the required 2% call blocking probability. By combining frequency planning, dynamic channel assignment and macro–micro cell hierarchy, an efficient and scalable cellular network can be designed for the university township.

TASK 4 – Analyse Best, Average and Worst-Case Behaviour 4 Marks

Guiding Enquiry

Where in the cell, and at what time of day, does each strategy perform best, typically, and worst — and why?

Investigate and interpret:

- Identify the position of the mobile (near the base station, mid-cell, cell corner) that yields best, average and worst SIR, and the geometry of interferers in each case.
- Identify the traffic condition (off-peak, average, busy-hour, event surge) that yields each blocking behaviour.
- analyse how shadowing and a changing path-loss exponent shift each case.
- Support each case with calculations, not just a table of values.

Evidence Expected

Justified best/average/worst analysis for SIR and blocking, supported by mathematical reasoning.

ANSWER

1. INTRODUCTION

The performance of a cellular network depends on the mobile user's position, traffic demand, signal strength, interference and environmental conditions. The Signal-to-Interference Ratio (SIR) changes according to the distance between the mobile device, serving base station and neighbouring co-channel base stations.

Best-case, average-case and worst-case analysis helps identify network performance under different conditions.

2. BEST-CASE PERFORMANCE

The best-case condition occurs when a mobile user is located close to the serving base station.

CHARACTERISTICS:

- The mobile user receives a strong signal from the serving base station.
- The distance to the serving base station is minimum.
- Signal strength is high and interference is comparatively low.
- Call quality and data transmission are better.
- The probability of call failure is low.

SIR ANALYSIS:

SIR = Desired signal power / Interference power

When the mobile user is near the serving base station, the desired signal power increases. Therefore, the SIR improves.

RESULT: Best-case performance provides strong signal reception, good call quality and reliable communication.

3. AVERAGE-CASE PERFORMANCE

The average-case condition occurs when a mobile user is located at a moderate distance from the serving base station.

CHARACTERISTICS:

- The received signal strength is moderate.
- Interference from neighbouring cells is present.
- Signal quality is generally acceptable.
- Call quality depends on traffic and environmental conditions.
- The network provides normal communication performance.

SIR ANALYSIS:

For a hexagonal cellular network with six first-tier co-channel interferers:

SIR = (D / R)^n / 6

Given:

- N = 7
- D / R = √(3 × 7) = 4.58
- n = 4

SIR = (4.58)^4 / 6= 73.5

SIR (dB) = 10 × log10(73.5)= 18.66 dB

RESULT: Average-case performance provides acceptable signal quality under the assumed ideal network conditions.

4. WORST-CASE PERFORMANCE

The worst-case condition occurs when a mobile user is located near the cell boundary or hexagonal cell corner.

CHARACTERISTICS:

- The distance from the serving base station is maximum.
- The desired signal strength becomes weak.
- Interference from neighbouring co-channel cells may increase.
- Call quality may decrease.
- Handoff failures and dropped calls may occur.

SIR ANALYSIS:

SIR = Desired signal power / Interference power

At the cell boundary, the desired signal becomes weaker because of the increased distance from the serving base station. Therefore, the SIR may decrease.

RESULT: Worst-case performance can cause poor call quality, increased interference and communication failure.

5. EFFECT OF PATH-LOSS EXPONENT ON SIR

The path-loss exponent determines how quickly signal strength decreases with distance.

SIR = (3N)^(n/2) / 6

Given N = 7:

SIR = 21^(n/2) / 6

The following values are obtained for different path-loss exponents:

Path-loss exponent (n)	Calculated SIR (linear)	SIR (dB)
2.5	7.49	8.75 dB
3	16.04	12.05 dB
4	73.50	18.66 dB

ANALYSIS:

- n = 2.5 represents a more open environment with slower signal reduction.
- n = 3 represents moderate signal attenuation.
- n = 4 represents a dense built-up environment with faster signal reduction.

In this simplified equal-distance interference model, a higher path-loss exponent increases the calculated SIR because interference from more distant co-channel cells decreases faster.

However, actual signal quality also depends on mobile position, buildings, obstacles and shadowing.

6. EFFECT OF TRAFFIC CONDITIONS

Traffic demand changes throughout the day. The busy hour is the period when the maximum number of users make calls.

LOW-TRAFFIC CONDITION:

- Fewer users make calls.
- More channels remain available.
- Call blocking probability is low.
- Network performance is generally better.

AVERAGE-TRAFFIC CONDITION:

- A moderate number of users access the network.
- Channels are shared among users.
- The network provides normal service.

HIGH-TRAFFIC CONDITION:

- A large number of users make calls simultaneously.
- Most available channels become occupied.
- Call blocking probability increases.
- Network congestion may occur.

The Erlang-B formula is used to calculate the blocking probability:

B(A,C) = (A^C / C!) / [Σ (A^k / k!), from k = 0 to C]

Where:

- B = Blocking probability
- A = Offered traffic in Erlangs
- C = Number of available channels

The target blocking probability for this assignment is 2%.

7. EFFECT OF SHADOWING AND ENVIRONMENT

Shadowing occurs when buildings, trees and other obstacles block or weaken radio signals.

EFFECTS OF SHADOWING:

- Signal strength decreases.
- Communication quality becomes unstable.
- Dead zones may occur.
- Call drops may increase.

- Additional base stations may be required in weak coverage areas.

In an open campus environment, signals can travel with fewer obstacles. In dense built-up areas, buildings and other structures can cause greater signal attenuation.

8. COMPARISON OF BEST, AVERAGE AND WORST CASES

Parameter	Best Case	Average Case	Worst Case
Mobile position	Near base station	Middle of cell	Cell boundary/corner
Signal strength	High	Moderate	Low
Interference impact	Low	Moderate	High
SIR	High	Moderate	Low
Call quality	Excellent	Acceptable	Poor
Call drop possibility	Low	Moderate	High

9. METHODS TO IMPROVE WORST-CASE PERFORMANCE

The following methods can improve network performance:

1. CELL SPLITTING: Divides large cells into smaller cells to improve coverage and capacity.
2. SECTORING: Uses directional antennas to reduce co-channel interference.
3. POWER CONTROL: Adjusts transmission power to maintain suitable signal strength.
4. HANDOFF OPTIMISATION: Transfers mobile users to a better serving base station.
5. BASE STATION PLACEMENT: Places base stations in weak signal areas to improve coverage.
6. DYNAMIC CHANNEL ASSIGNMENT: Allocates available channels according to traffic demand.

10. CONCLUSION

Best-case, average-case and worst-case analysis helps evaluate cellular network performance under different conditions. Users near the base station generally receive stronger signals, while users near cell boundaries may experience weaker signals and higher interference. Traffic congestion and shadowing also affect call quality and blocking probability. By using sectoring, cell splitting, power control and proper base station placement, the network can improve coverage, reduce interference and maintain the required 2% call blocking probability.

TASK 5 – Scaling and Sensitivity Analysis **3 Marks**

Guiding Enquiry

How do capacity, interference and the number of cells scale as demand and the environment change, and at what point does each strategy stop being practical? Investigate and interpret:

- Express proportionalities (e.g., capacity $\propto 1/N$, SIR $\propto N^{(n/2)}$, cells $\propto 1/R^2$) and compare growth of the strategies.
- Interpret what happens when subscribers grow from 1,000 to 100,000+ and when n varies from 2.5 to 4.
- Identify a break-even scale beyond which a different strategy becomes preferable.

Evidence Expected

Correct scaling relations and an interpretation tied to the township application.

ANSWER

1. INTRODUCTION

Scaling and sensitivity analysis are used to study how changes in network parameters affect cellular network capacity, coverage and interference. The important parameters include cluster size, cell radius, path-loss exponent, available channels and number of subscribers.

The university township network must support subscriber growth from 1,000 to 100,000 or more without increasing the available spectrum of 336 duplex channels.

2. EFFECT OF CLUSTER SIZE ON NETWORK CAPACITY

The number of channels allocated to each cell depends on the cluster size.

Channels per cell = Total available channels / Cluster size

$C = S / N$

Given:

$S = 336$ channels

For different cluster sizes:

Cluster size (N)	Channels per cell
3	112
4	84
7	48
12	28

ANALYSIS:

- A smaller cluster size provides more channels per cell.
- A larger cluster size provides fewer channels per cell.
- A larger cluster size increases frequency reuse distance and generally improves SIR.
- A smaller cluster size increases capacity but may increase co-channel interference. Therefore, cluster size must be selected by balancing capacity and interference.

3. EFFECT OF CELL RADIUS ON NETWORK CAPACITY

The area of one hexagonal cell is:

$Ac = (3 \times \sqrt{3} / 2) \times R^2$

$Ac = 2.598 \times R^2$

The number of cells required to cover the township is:

M = Total area / Area of one cell

$M = 12 / (2.598 \times R^2)$

Therefore:

Number of cells is inversely proportional to the square of cell radius.

$M \propto 1 / R^2$

When the cell radius decreases, more cells are required to cover the same area.

EXAMPLE:

Cell radius (R)	Approximate number of cells
1 km	5
0.5 km	19
0.25 km	74

ANALYSIS:

Smaller cells increase spatial frequency reuse and network capacity. However, they require more base stations, installation costs and handoffs.

4. EFFECT OF PATH-LOSS EXPONENT ON SIR

The path-loss exponent determines how quickly the signal strength decreases with distance.

SIR = (3N)^(n/2) / 6

For N = 7:

SIR = 21^(n/2) / 6

The calculated values are:

Path-loss exponent (n)	SIR (linear)	SIR (dB)
2.5	7.49	8.75 dB
3	16.04	12.05 dB
4	73.50	18.66 dB

ANALYSIS:

- A path-loss exponent of 2.5 represents a relatively open environment.
- A path-loss exponent of 3 represents moderate signal attenuation.
- A path-loss exponent of 4 represents a dense built-up environment.

In this simplified model, increasing the path-loss exponent improves the calculated SIR because signals from distant interfering cells decrease faster. Actual performance also depends on obstacles, shadowing and mobile position.

5. EFFECT OF SUBSCRIBER GROWTH

The busy-hour traffic per subscriber is 0.025 Erlang.

Total traffic demand = Number of subscribers × Traffic per subscriber

The estimated traffic demand is:

Subscribers	Total traffic demand
1,000	25 Erlangs
10,000	250 Erlangs
20,000	500 Erlangs
100,000	2,500 Erlangs

From Task 3, one cell supports approximately 38.39 Erlangs at 2% blocking probability, assuming 48 channels per cell.

Required cells = Total traffic demand / Traffic capacity per cell

Subscribers	Required cells (approx.)
1,000	1
10,000	7
20,000	14
100,000	66

ANALYSIS:

As the number of subscribers increases, traffic demand also increases. Therefore, the network requires more cells, better channel allocation and improved interference management.

6. SENSITIVITY ANALYSIS

Sensitivity analysis identifies how changes in one parameter affect network performance.

Parameter changed	Effect on network
Cluster size increases	Channels per cell decrease
Cluster size decreases	Channels per cell increase, but interference may increase
Cell radius increases	Number of cells decreases
Cell radius decreases	Number of cells and spatial reuse increase
Path-loss exponent increases	Idealized SIR increases for distant interferers
Subscriber count increases	Traffic demand and blocking risk increase
Sectoring is introduced	Co-channel interference is reduced

7. CAPACITY AND SCALING RELATIONSHIPS

The important mathematical relationships are:

1. Channels per cell:

$C = S / N$

Capacity per cell is inversely proportional to cluster size.

2. Frequency reuse distance:

$D = R \times \sqrt{(3N)}$

Reuse distance increases with cell radius and cluster size.

3. Number of cells:

$M = A / (2.598 \times R^2)$

Number of cells is inversely proportional to the square of cell radius.

4. Signal-to-interference ratio:

SIR = (3N)^(n/2) / 6

In the idealized model, SIR increases with cluster size and path-loss exponent.

5. Total traffic demand:

A_T = Number of subscribers × 0.025

Traffic demand increases directly with subscriber count.

8. BREAK-EVEN ANALYSIS

Break-even analysis identifies the cell radius required to support a given number of subscribers using the available channels.

Using the assumptions from Task 3:

- Township area = 12 km²
- Traffic capacity per cell = 38.39 Erlangs
- Cell area = 2.598 × R²
- Traffic per subscriber = 0.025 Erlang

Required traffic capacity = Number of subscribers × 0.025

Required number of cells = Required traffic capacity / 38.39

Using the cell-area equation, the approximate maximum cell radius is:

R = √[12 × 38.39 / (2.598 × Required traffic capacity)]

EXAMPLE FOR 10,000 SUBSCRIBERS:

Required traffic = 10,000 × 0.025

Required traffic = 250 Erlangs

Required cells = 250 / 38.39

Required cells ≈ 7

Required cell radius:

R = √[12 × 38.39 / (2.598 × 250)]

R ≈ 0.84 km

Therefore, the cell radius should be approximately 0.84 km or smaller under these simplified assumptions to support 10,000 subscribers.

The same calculation gives approximately:

Subscribers	Approximate maximum cell radius
10,000	0.84 km
20,000	0.60 km
100,000	0.27 km

Note: These are idealized estimates based on the assumed per-cell capacity. Actual deployment must also consider complete frequency reuse clusters, cell placement, interference, coverage and base-station limitations.

9. CONCLUSION

The scaling and sensitivity analysis shows that network capacity depends on cluster size, cell radius, path-loss exponent and subscriber traffic demand. Reducing cell radius increases the number of cells and improves spatial frequency reuse, while increasing cluster size improves the idealized SIR but reduces channels per cell. As subscriber numbers increase from 1,000 to 100,000 or more, the network requires cell splitting, sectoring and efficient channel allocation. These techniques help improve capacity and maintain the required 2% call blocking probability without increasing the available spectrum.

TASK 6 – Formulate the Recurrence for Repeated Cell Splitting 3 Marks

Guiding Enquiry

If overloaded cells are split repeatedly, how can the number of cells, capacity, transmit power and handoff load after k rounds be expressed in terms of round $k - 1$?

Investigate and interpret:

- State the original problem size (cells and demand) and the subproblem produced by each split (radius $R \rightarrow R/2$).
- Identify the number of new cells per split, the work/cost per round, and the base case (the initial layout).
- Formulate recurrences for number of cells, offered capacity and required transmit power ($P' = P/2^n$), derived from your own splitting procedure.
- Allow for selective splitting (only a fraction of cells is overloaded) as well as uniform splitting.

Evidence Expected

Correct recurrence(s) with clearly identified components and base case.

ANSWER

1. INTRODUCTION

Cell splitting is a network expansion technique used to increase cellular network capacity without increasing the available radio spectrum. In this method, a large cell is divided into smaller cells by installing additional base stations.

As the number of subscribers increases from 1,000 to 100,000 or more, cell splitting helps the network handle additional traffic while maintaining communication quality and reducing congestion.

2. CONCEPT OF CELL SPLITTING

Cell splitting divides an existing cell into four smaller cells.

- Original cell radius = R
- New cell radius = R/2

- Number of new cells = 4
- Coverage area remains approximately the same.
- More base stations are required.
- Frequency reuse opportunities increase.

CELL RADIUS AFTER SPLITTING

New radius = R / 2

If the original radius is 1 km:

New radius = 1 / 2

New radius = 0.5 km

After another splitting operation:

New radius = 0.5 / 2

New radius = 0.25 km

3. RECURRENCE RELATION FOR NUMBER OF CELLS

In each splitting stage, one cell is replaced by four smaller cells.

Therefore:

Number of cells after splitting = 4 × Previous number of cells

Let C(k) represent the number of cells after k splitting stages.

C(k) = 4 × C(k-1)

Where:

- C(k) = Number of cells at the current stage.
- C(k-1) = Number of cells at the previous stage.
- k = Splitting stage.

BASE CASE:

C(0) = 1

RECURRENCE RELATION:

C(k) = 4C(k-1)

4. RECURRENCE RELATION FOR CELL RADIUS

During each splitting stage, the cell radius becomes half of its previous value.

Let R(k) represent the cell radius after k stages.

R(k) = R(k-1) / 2

BASE CASE:

R(0) = R

RECURRENCE RELATION:

R(k) = R(k-1) / 2

The resulting radius after k stages is:

R(k) = R / 2^k

5. RECURRENCE RELATION FOR NETWORK CAPACITY

Cell splitting increases the number of cells and provides more opportunities for frequency reuse.

Assuming each smaller cell can support the same traffic capacity as the original cell under the same channel allocation:

T(k) = 4 × T(k-1)

Where:

- T(k) = Total network traffic capacity after k stages.
- T(k-1) = Previous traffic capacity.

BASE CASE:

$T(0) = T_0$

RECURRENCE RELATION:

T(k) = 4T(k-1)

Therefore:

$T(k) = 4^k \times T_0$

Note: This is an idealized capacity model. Actual capacity depends on channel allocation, interference, traffic distribution and frequency reuse planning.

6. RECURRENCE RELATION FOR TRANSMISSION POWER

When the cell radius is reduced, the base station transmission power can also be reduced to maintain approximately the same received signal level at the cell boundary.

Using the path-loss model:

Received power ∝ Transmitted power / R^n

For the same received power:

P(k) = P(k-1) / 2^n

Where:

- P(k) = Transmission power after splitting.
- P(k-1) = Previous transmission power.
- n = Path-loss exponent.

For n = 4:

P(k) = P(k-1) / 16

Therefore, the transmission power becomes approximately one-sixteenth of its previous value after each splitting stage, under the stated propagation assumptions.

7. UNIFORM AND SELECTIVE CELL SPLITTING

A. UNIFORM CELL SPLITTING

Uniform splitting divides all cells into smaller cells.

Characteristics:

- All cells undergo splitting.
- Coverage becomes more detailed.
- Network capacity increases.
- More base stations are required.
- Installation and maintenance costs increase.

B. SELECTIVE CELL SPLITTING

Selective splitting divides only cells with high traffic demand.

Characteristics:

- Only congested cells are divided.
- Low-traffic cells remain unchanged.
- Infrastructure cost is reduced.
- Network resources are used efficiently.
- It is suitable for crowded university locations.

Examples include classrooms, libraries, canteens and stadium areas.

8. COMPARISON OF CELL SPLITTING STAGES

Assume the original cell radius is 1 km.

Splitting stage	Number of cells	Cell radius
0	1	1 km
1	4	0.5 km
2	16	0.25 km
3	64	0.125 km
4	256	0.0625 km

The table shows that repeated cell splitting increases the number of cells while reducing the cell radius.

9. WORK AND COST ANALYSIS

Cell splitting increases the number of base stations and network operations.

Total installation cost = Number of new base stations × Cost per base station

$C = M \times Cb$

Where:

- C = Installation cost
- M = Number of base stations
- Cb = Cost per base station

Additional costs include:

1. Base station installation.
2. Antenna and equipment maintenance.
3. Power consumption.
4. Handoff management.
5. Network monitoring and interference control.

Selective cell splitting reduces unnecessary infrastructure expenses by concentrating resources in high-demand areas.

10. CONCLUSION

Cell splitting is an effective technique for increasing cellular network capacity without increasing the available spectrum. The recurrence relations show that the number of cells increases by a factor of four while the cell radius becomes half after each splitting stage. Transmission power can also be adjusted according to the path-loss exponent. Uniform splitting provides wider capacity improvement, whereas selective splitting is more economical for congested locations. Therefore, selective cell splitting is suitable for supporting the growing subscriber population in a university township.

TASK 7 – Solve the Recurrence and Interpret the Result **4 Marks**

Guiding Enquiry

How many splitting rounds are needed to carry the forecast demand, and what do the closed-form results say about cost and mobility? Investigate and interpret:

- Solve using iteration/substitution (or another suitable method) and justify the method.
- Derive the minimum number of rounds $k = \lceil \log_4(D/C_0) \rceil$ for uniform splitting and its selective counterpart.
- Interpret the result: growth in sites, reduction in transmit power, and the increase in handoff rate as radius shrinks.

Evidence Expected

Complete solution with method justification and an interpretation of the closed form.

ANSWER

TASK 7 – SOLVE THE RECURRENCE RELATIONS USING ITERATION AND SUBSTITUTION

1. INTRODUCTION

Recurrence relations are used to calculate the number of cells, cell radius and network capacity after repeated cell splitting. By solving these relations, we can estimate how many splitting stages are required to support increasing subscriber demand.

Cell splitting increases network capacity by dividing large cells into smaller cells without increasing the available spectrum.

2. RECURRENCE RELATION FOR NUMBER OF CELLS

The recurrence relation is:

$C(k) = 4 \times C(k-1)$

Base case:

$C(0) = 1$

Using iteration:

$C(1) = 4 \times 1 = 4$

$C(2) = 4 \times 4 = 16$

$C(3) = 4 \times 16 = 64$

C(4) = 4 × 64 = 256

Therefore, the general solution is:

C(k) = 4^k

Where k represents the number of splitting stages.

3. RECURRENCE RELATION FOR CELL RADIUS

The cell radius becomes half after every splitting stage.

R(k) = R(k-1) / 2

Base case:

R(0) = 1 km

Using iteration:

R(1) = 1 / 2 = 0.5 km

R(2) = 0.5 / 2 = 0.25 km

R(3) = 0.25 / 2 = 0.125 km

R(4) = 0.125 / 2 = 0.0625 km

General solution:

R(k) = R(0) / 2^k

4. RECURRENCE RELATION FOR NETWORK CAPACITY

Assuming each new cell provides the same traffic capacity as the original cell:

T(k) = 4 × T(k-1)

Base case:

$T(0) = T_0$

Using iteration:

$T(1) = 4T_0$

$T(2) = 16T_0$

$T(3) = 64T_0$

Therefore, the general solution is:

$T(k) = 4^k \times T_0$

This represents the ideal increase in total capacity after repeated splitting.

5. CALCULATION OF REQUIRED SPLITTING STAGES

From Task 3, the estimated capacity of one cell is:

$T_0 = 38.39$ Erlangs

For 100,000 subscribers:

Traffic demand = 100,000 × 0.025

Traffic demand = 2,500 Erlangs

The required number of ideal cell-capacity units is:

Required cells = 2,500 / 38.39

Required cells = 65.12

Rounding up:

Required cells = 66

For uniform four-way splitting:

4^k ≥ 66

Taking logarithms:

$k \geq \log_4(66)$

k ≥ 3.02

Therefore:

k = 4 splitting stages

At four stages:

Number of cells = 4^4 = 256 cells

The ideal capacity becomes:

Total capacity = 256 × 38.39

Total capacity = 9,827.84 Erlangs

6. SELECTIVE CELL SPLITTING

Selective cell splitting divides only cells experiencing high traffic demand.

The required number of additional capacity units is:

Additional cells = Required cells − Existing cells

For example:

Required cells = 66

Existing cells = 5

Additional cells = 66 − 5

Additional cells = 61

Therefore, selective splitting focuses on congested locations instead of splitting every cell. It reduces unnecessary installation costs and improves resource utilisation.

7. TRANSMISSION POWER AFTER SPLITTING

The transmission power can be reduced when the cell radius is reduced.

P(k) = P(k-1) / 2^n

Assume:

- Initial transmission power = 40 W
- Path-loss exponent = 4
- Splitting stages = 1

P(1) = 40 / 2^4

P(1) = 40 / 16

P(1) = 2.5 W

After the second splitting stage:

P(2) = 2.5 / 16

P(2) = 0.15625 W

This is an idealized power calculation for maintaining the same received signal level at the cell boundary.

8. COMPARISON OF UNIFORM AND SELECTIVE SPLITTING

Parameter	Uniform splitting	Selective splitting
Splitting method	All cells	High-traffic cells
Infrastructure	More	Less
Cost	High	Lower
Capacity improvement	Across the network	In selected areas
Resource utilisation	May be inefficient	More efficient
Suitable for	Uniform traffic	Uneven traffic

9. CONCLUSION

The recurrence relations were solved using iteration and substitution. The calculations show that four-way cell splitting increases the number of cells by a factor of four at each stage, while the cell radius becomes half. For the assumed 100,000-subscriber demand, four uniform splitting stages provide 256 ideal cell units. However, selective splitting is more economical because it focuses on congested locations. Proper cell planning, power control and channel allocation are necessary to achieve efficient network performance.

Note: The capacity calculation assumes each new cell receives an independent equivalent channel allocation. Actual capacity must be verified using the fixed 336-channel spectrum, frequency reuse planning and interference constraints.

TASK 8 – Implement and Test the Planning Strategies **4 Marks**

Guiding Enquiry

Can a simulator built from your own strategies reproduce the behaviour predicted by your mathematics?

Investigate and interpret:

- Implement all strategies (e.g., Python/NumPy): hex-grid generator, channel assignment, Monte-Carlo SIR evaluator and Erlang-B calculator.
- Use scaled datasets: 1,000 → 10,000 → 20,000 → 100,000 subscribers with the same random seeds across strategies.
- Include test cases: N = 3, 4, 7, 12; n = 2.5, 3, 4; sectored and unsectored layouts.
- Sanity-check with a known case (N = 7, n = 4 under the six-equidistant-interferer approximation gives about 18.7 dB).

Evidence Expected

Working code, reproducible procedure and correctness checks aligned to the analysed strategies.

ANSWER

TASK 8 – IMPLEMENT AND TEST CELLULAR NETWORK DESIGN STRATEGIES USING PYTHON

1. INTRODUCTION

Python is used to implement and test different cellular network design strategies. The simulation helps analyse frequency reuse, channel allocation, signal-to-interference ratio (SIR), traffic capacity and call blocking probability.

The main objective is to compare different cluster sizes and path-loss exponents and identify a suitable network configuration for the university township.

2. SOFTWARE AND TOOLS USED

- Programming Language: Python
- IDE: Visual Studio Code
- Libraries: NumPy, Matplotlib
- Simulation Method: Numerical simulation and Erlang-B calculation

3. IMPLEMENTATION OBJECTIVES

1. Generate different frequency reuse cluster sizes.
2. Calculate channels allocated per cell.
3. Calculate frequency reuse distance.
4. Estimate signal-to-interference ratio.
5. Calculate traffic capacity using the Erlang-B model.
6. Compare different path-loss exponents.
7. Analyse the performance of sectorised and unsectorised networks.

4. PYTHON PROGRAM

The following program calculates channel allocation, reuse distance, SIR, Erlang-B traffic capacity and Monte Carlo SIR performance for different network configurations.

CODE:

```
import numpy as np

# Network parameters
S = 336
R = 1
N_values = [3, 4, 7, 12]
n_values = [2.5, 3, 4]
users = 5000
blocking_limit = 0.02

# Erlang-B formula
def erlang_b(A, C):
    B = 1.0
    for i in range(1, C + 1):
        B = (A * B) / (i + A * B)
    return B

# Find traffic capacity at 2% blocking
def traffic_capacity(C):
    low, high = 0, C * 2

    for _ in range(100):
        mid = (low + high) / 2
```

```
            if erlang_b(mid, C) > blocking_limit:
                high = mid
            else:
                low = mid

    return (low + high) / 2

    # Monte Carlo SIR simulation
    def simulate_sir(N, n, sectoring=False):
        D = R * np.sqrt(3 * N)

        # Random mobile positions inside a circular cell
        radius = R * np.sqrt(np.random.random(users))
        angle = 2 * np.pi * np.random.random(users)

        x = radius * np.cos(angle)
        y = radius * np.sin(angle)

        # Six first-tier co-channel base stations
        angles = np.arange(6) * np.pi / 3

        interference = np.zeros(users)

        count = 2 if sectoring else 6

        for a in angles[:count]:
        ix = D * np.cos(a)
        iy = D * np.sin(a)

            distance = np.sqrt((x - ix)**2 + (y - iy)**2)
            interference += distance ** (-n)

        desired_distance = np.maximum(radius, 0.01)
        signal = desired_distance ** (-n)
```

```
    sir = signal / interference
    sir_db = 10 * np.log10(sir)

    return np.percentile(sir_db, 5), np.mean(sir_db)

# Main simulation
np.random.seed(42)

for N in N_values:
    channels = S // N
    D = R * np.sqrt(3 * N)
    capacity = traffic_capacity(channels)

    print("\nCluster Size:", N)
    print("Channels per cell:", channels)
    print("Reuse distance:", round(D, 2), "km")
    print("Traffic capacity:", round(capacity, 2), "Erlangs")

    for n in n_values:
        p5, avg = simulate_sir(N, n)

        p5_sector, avg_sector = simulate_sir(
            N, n, sectoring=True
        )

        print("Path-loss exponent:", n)
        print("Unsectorised 5th percentile SIR:",
              round(p5, 2), "dB")
        print("Unsectorised average SIR:",
              round(avg, 2), "dB")
        print("Sectorised 5th percentile SIR:",
              round(p5_sector, 2), "dB")
        print("Sectorised average SIR:",
              round(avg_sector, 2), "dB")
```

5. INPUT DATA

The program uses the following input values:

Parameter	Values
Available channels	336
Cell radius	1 km
Cluster sizes	3, 4, 7, 12
Path-loss exponent	2.5, 3, 4
Monte Carlo samples	5,000
Blocking probability	2%
Interferers without sectoring	6
Interferers with sectoring	2 (idealized assumption)

6. OUTPUT

The program displays:

- Cluster size.
- Number of channels per cell.
- Frequency reuse distance.
- Traffic capacity at 2% blocking.
- Average simulated SIR.
- 5th percentile SIR.
- Sectorised and unsectorised SIR results.

EXPECTED BASIC OUTPUT FORMAT:

Cluster Size: 3

Channels per cell: 112

Reuse distance: 3.0 km

Traffic capacity: [calculated value] Erlangs

Path-loss exponent: 2.5

Unsectorised 5th percentile SIR: [value] dB

Unsectorised average SIR: [value] dB

Sectorised 5th percentile SIR: [value] dB

Sectorised average SIR: [value] dB

The remaining values are calculated by Python when the program is executed. Actual Monte Carlo results may vary if the random seed is changed.

```
PS C:\Users\dhara\OneDrive\ドキュメント\ITA033
ms\Python\Python313\python.exe' 'c:\Users\dhar
'--' 'c:\Users\dhara\OneDrive\ドキュメント\ITA

Cluster Size: 3
Channels per cell: 112
Reuse distance: 3.0 km
Traffic capacity: 99.62 Erlangs
Path-loss exponent: 2.5
Unsectorised 5th percentile SIR: 3.66 dB
Unsectorised average SIR: 9.25 dB
Sectorised 5th percentile SIR: 7.08 dB
Sectorised average SIR: 14.29 dB
Path-loss exponent: 3
Unsectorised 5th percentile SIR: 5.74 dB
Unsectorised average SIR: 12.3 dB
Sectorised 5th percentile SIR: 9.2 dB
Sectorised average SIR: 17.63 dB
Path-loss exponent: 4
Unsectorised 5th percentile SIR: 9.84 dB
Unsectorised average SIR: 18.97 dB
Sectorised 5th percentile SIR: 13.02 dB
Sectorised average SIR: 24.47 dB
```

7. TESTING AND VALIDATION

The following configurations are tested:

Test case	Cluster size	Path-loss exponent
Test 1	3	2.5
Test 2	4	3
Test 3	7	4
Test 4	12	4

The program is tested to verify that:

1. Channels per cell are calculated correctly.
2. Reuse distance increases with cluster size.
3. Traffic capacity is calculated at 2% blocking.
4. SIR changes according to mobile position and path-loss exponent.
5. Sectorisation is compared with unsectorised operation.

8. SANITY CHECK

For cluster size N = 7 and path-loss exponent n = 4:

SIR = (3N)^(n/2) / 6

SIR = 21^2 / 6

SIR = 73.5

SIR in dB = 10 × log10(73.5)

SIR = 18.66 dB ≈ 18.7 dB

This matches the expected theoretical value for the idealized six-interferer model.

Note: The Monte Carlo simulation uses a circular approximation of the cell and six equally spaced interfering base stations. Its results are not expected to match the theoretical 18.7 dB exactly. The sectorised simulation uses two interferers as a simplified assumption.

9. RESULT

The cellular network simulation was implemented successfully using Python and NumPy. Different cluster sizes and path-loss exponents were tested to evaluate channel allocation, traffic capacity and signal-to-interference ratio. The results help compare sectorised and unsectorised networks and identify suitable parameters for efficient cellular network planning.

TASK 9 – Record and Present Experimental Performance 3 Marks

Guiding Enquiry

Which measurements best expose the differences between your strategies, and how should they be presented?

Investigate and interpret:

- Record SIR distribution (mean, 5th percentile), outage probability, blocking probability, channels per cell, sites required and run-time.
- Present tables and plots such as SIR CDFs, blocking vs offered load and capacity vs N.
- Ensure the presentation allows direct comparison between strategies.

Evidence Expected

Systematically collected data presented in comparable tables and graphs.

ANSWER

1. INTRODUCTION

Experimental performance evaluation is used to compare different cellular network design strategies through simulation results. It helps identify how cluster size, cell radius, path-loss exponent and sectoring affect network capacity, interference and call quality. The main objective is to determine which strategy provides better network performance while maintaining the required 2% call blocking probability.

2. PERFORMANCE PARAMETERS

The following parameters are used to evaluate the cellular network:

1. SIGNAL-TO-INTERFERENCE RATIO (SIR): Measures the strength of the desired signal compared with interference.
2. 5TH PERCENTILE SIR: Represents the SIR value below which 5% of simulated users fall.
3. OUTAGE PROBABILITY: Percentage of users whose SIR is below the minimum acceptable SIR.
4. CALL BLOCKING PROBABILITY: Percentage of call attempts rejected because channels are unavailable.
5. CHANNELS PER CELL: Number of voice channels allocated to each cell.
6. NUMBER OF BASE STATIONS: Indicates the infrastructure required to cover the

township.

7. SIMULATION TIME: Time required to complete the network simulation.

3. EXPERIMENTAL CONFIGURATION

The following values are used for simulation:

Parameter	Value
Township area	12 km²
Available duplex channels	336
Cell radius	1 km
Cluster sizes	3, 4, 7, 12
Path-loss exponent	2.5, 3, 4
Blocking probability target	2%
Monte Carlo samples	5,000
Minimum acceptable SIR	12–18 dB

4. EXPERIMENTAL PERFORMANCE TABLE

The following table shows the theoretical performance comparison for different cluster sizes at a path-loss exponent of 4.

SIR = $(3N)^2 / 6$

Cluster size	Channels per cell	Reuse distance (km)	SIR (dB)
3	112	3.00	11.76
4	84	3.46	13.80
7	48	4.58	18.66
12	28	6.00	23.80

ANALYSIS:

- Cluster size 3 provides more channels per cell but lower SIR.
- Cluster size 4 provides moderate capacity and interference performance.
- Cluster size 7 provides a balance between channel availability and SIR.
- Cluster size 12 provides higher SIR but fewer channels per cell.

5. EFFECT OF SECTORING

Sectoring uses directional antennas to reduce interference from neighbouring co-channel cells.

For an idealized 120-degree sectoring model with two dominant interferers:

SIR = $(3N)^2 / 2$

For N = 7:

SIR = $21^2 / 2$

SIR = 220.5

SIR (dB) = 10 × log10(220.5)

SIR = 23.43 dB

Without sectoring, SIR = 18.66 dB.

With sectoring, SIR = 23.43 dB.

Therefore, sectoring improves the theoretical SIR by approximately 4.77 dB under this simplified assumption.

6. OUTAGE PROBABILITY

Outage probability measures the percentage of users receiving an SIR below the minimum acceptable threshold.

Outage probability = (Number of users below SIR threshold / Total simulated users) × 100

For example:

If 250 users out of 5,000 users experience SIR below the required threshold:

Outage probability = (250 / 5,000) × 100

Outage probability = 5%

A lower outage probability indicates better network coverage and communication quality.

7. CALL BLOCKING PROBABILITY

Call blocking probability measures the percentage of call attempts that cannot be served because all available channels are occupied.

Blocking probability = (Blocked calls / Total call attempts) × 100

The network target is:

Blocking probability ≤ 2%

The Erlang-B model is used to estimate the traffic capacity required to maintain this target.

8. COMPARISON OF NETWORK STRATEGIES

Strategy	Main advantage	Limitation
Fixed frequency reuse	Simple channel planning	Fixed channel allocation
Sectorised reuse	Reduces interference	Requires directional antennas
Dynamic channel assignment	Improves channel utilisation	More complex allocation
Macro–micro hierarchy	Improves capacity in crowded areas	Higher infrastructure cost

9. GRAPHICAL REPRESENTATION

The following graphs can be generated using Python and Matplotlib:

GRAPH 1 – CLUSTER SIZE VS SIR

- X-axis: Cluster size

- Y-axis: SIR in dB
- Purpose: Shows how SIR increases with cluster size.

GRAPH 2 – CLUSTER SIZE VS CHANNELS PER CELL

- X-axis: Cluster size
- Y-axis: Channels per cell
- Purpose: Shows how available channels per cell decrease when cluster size increases.

GRAPH 3 – PATH-LOSS EXPONENT VS SIR

- X-axis: Path-loss exponent
- Y-axis: SIR in dB
- Purpose: Shows the effect of environmental signal

attenuation. GRAPH 4 – SECTORISED VS UNSECTORISED SIR

- X-axis: Network configuration
- Y-axis: SIR in dB
- Purpose: Compares interference performance with and without sectoring.

10. RESULT ANALYSIS

The experimental comparison shows that cluster size has a major effect on network capacity and interference. Smaller cluster sizes provide more channels per cell, while larger cluster sizes improve frequency reuse distance and SIR.

Sectoring improves interference performance by reducing the number of dominant co-channel interferers. Dynamic channel assignment improves channel utilisation, while macro–micro cell hierarchy supports higher traffic demand in crowded locations.

The final selection should consider both capacity and interference rather than SIR alone.

11. CONCLUSION

Experimental performance evaluation helps identify the most suitable cellular network design strategy. The comparison shows that cluster size 7 provides a reasonable balance between channel allocation and interference performance under the given assumptions. Sectoring can further improve SIR, while dynamic channel assignment and cell splitting help support increasing subscriber demand. These techniques can improve network efficiency and maintain the required 2% call blocking probability

TASK 10 – Compare Theory with Simulation **4**

Marks Guiding Enquiry

Where do the simulated results agree with your analytical predictions, where do they differ, and what analyses the gap?

Investigate and interpret:

- Compare analytical and simulated SIR and blocking for each strategy.
- Identify sources of deviation: equidistant-interferer approximation, shadowing, finite tiers, mobile-position distribution, Poisson-arrival and infinite-source assumptions in Erlang-B, integer rounding of N and channels.
- Decide which strategy scales best and defend this using both results.

Evidence Expected

A reasoned comparison of theory and simulation with analyseed discrepancies.

TASK 10 – COMPARE THEORETICAL RESULTS WITH SIMULATION RESULTS

1. INTRODUCTION

Theoretical analysis and computer simulation are important methods for evaluating cellular network performance. Theoretical calculations use mathematical formulas to predict network capacity, frequency reuse distance and signal-to-interference ratio (SIR). Simulation uses computer programs to study network behaviour under different conditions.

The main objective is to compare both results, identify differences and determine the most suitable cellular network design strategy.

2. THEORETICAL ANALYSIS

Theoretical analysis uses mathematical equations to calculate network parameters.

The important equations are:

A. FREQUENCY REUSE DISTANCE

$D = R \times \sqrt{(3N)}$

B. CHANNELS PER CELL

$C = S / N$

C. HEXAGONAL CELL AREA

Ac = 2.598 × R^2

D. NUMBER OF CELLS

M = Total area / Area of one cell

E. SIGNAL-TO-INTERFERENCE RATIO

SIR = (3N)^(n/2) / 6

These equations provide estimated network performance under ideal conditions.

3. SIMULATION ANALYSIS

Simulation is performed using Python and NumPy to analyse different network configurations.

The simulation considers:

- Different cluster sizes: 3, 4, 7 and 12.
- Different path-loss exponents: 2.5, 3 and 4.
- Random mobile user positions.
- Co-channel interference.
- Sectorised and unsectorised configurations.
- Traffic capacity and call blocking probability.

Simulation helps analyse network performance under varying conditions.

4. COMPARISON OF THEORETICAL AND SIMULATION RESULTS

The following table shows the theoretical reference values for a cluster size of 7 and path-loss exponent of 4, together with the simulation comparison criteria.

Parameter	Theoretical result	Simulation result
Cluster size	7	7
Channels per cell	48	48
Reuse distance	4.58 km	Approximately 4.58 km
SIR	18.66 dB	Calculated using random user positions
Blocking probability	2% target	Calculated using Erlang-B
Interference	Idealized	Position-dependent

The exact simulation SIR depends on the random mobile positions and the distances to interfering base stations. Therefore, it may differ from the theoretical value.

5. REASONS FOR DIFFERENCES BETWEEN THEORY AND SIMULATION

1. MOBILE USER POSITION

Theoretical calculations assume simplified distances between base stations and mobile

users. In simulation, mobile users occupy different positions, causing variations in signal strength.

2. INTERFERENCE APPROXIMATION

The theoretical model assumes six equal-distance first-tier interferers. Actual interference depends on the distance and strength of each interfering signal.

3. SHADOWING EFFECT

Buildings, trees and other obstacles weaken radio signals. These effects may not be included in the basic theoretical equations.

4. PATH-LOSS VARIATION

Different environments have different path-loss exponents. Open campus areas and dense buildings produce different signal propagation conditions.

5. FINITE NETWORK SIZE

The theoretical model may assume an infinite hexagonal network. Simulation uses a limited number of cells, which changes the interference pattern.

6. TRAFFIC DISTRIBUTION

Theoretical Erlang-B calculations assume a statistical traffic model. Actual traffic may vary according to location and time.

6. THEORY VS SIMULATION: ADVANTAGES

Theoretical analysis	Simulation analysis
Uses mathematical equations	Uses computer programs
Provides ideal estimates	Represents varying conditions
Easy to calculate	Can handle complex scenarios
Requires fewer resources	Requires more computation
Limited environmental details	Can include user positions and interference

7. SELECTION OF THE BEST NETWORK STRATEGY

Based on the theoretical and simulation analysis:

- Fixed frequency reuse provides simple and predictable channel allocation.
- Sectorised reuse reduces co-channel interference.
- Dynamic channel assignment improves channel utilisation.
- Macro–micro cell hierarchy increases capacity in crowded areas.

RECOMMENDED APPROACH:

A combination of frequency reuse, sectoring, dynamic channel assignment and selective cell splitting is suitable for the university township.

This approach provides better capacity, coverage and interference management without increasing the available spectrum.

8. CONCLUSION

Theoretical analysis provides a mathematical foundation for cellular network design, while simulation helps evaluate actual network behaviour under different conditions. Differences between the two methods occur because of mobile user positions, interference, shadowing, path-loss variation and traffic distribution. Combining theoretical calculations with simulation results provides a more reliable network design. Sectoring, dynamic channel assignment and selective cell splitting are effective techniques for improving network performance and supporting future subscriber growth.

TASK 11 – Recommend and Defend the Final Design **4**

Marks Guiding Enquiry

Which design would you actually deploy in the township, and what evidence makes it better than the alternatives?

Investigate and interpret:

- State the recommended strategy, parameters (N, R, sectoring) and expected capacity/SIR/blocking.
- analyse why it is superior to the alternatives using your analysis and simulation.
- interpret planning cost, trade-offs (trunking efficiency vs interference, handoff load vs capacity) and limitations.

- Judge whether the design remains practical as the township grows past 100,000 subscribers.

Evidence Expected

A technically justified recommendation supported by analysis and simulation; choosing the highest-capacity option alone is not sufficient.

ANSWER

TASK 11 – RECOMMEND THE FINAL CELLULAR NETWORK DESIGN

1. INTRODUCTION

The final cellular network design is selected by considering network capacity, frequency reuse, interference, coverage, infrastructure cost and future subscriber growth.

The university township has an area of 12 km² and must support more than 100,000 subscribers without increasing the available spectrum of 336 duplex voice channels.

The recommended design combines frequency reuse, sectoring, dynamic channel assignment and selective cell splitting.

2. RECOMMENDED NETWORK PARAMETERS

The following parameters are selected for the final network:

Parameter	Recommended value
Township area	12 km²
Available duplex channels	336
Cluster size (N)	12
Cell radius (R)	Approximately 0.187 km
Channels per cell	28
Path-loss exponent	2.5–4
Minimum SIR target	12 dB
Blocking probability	Maximum 2%
Sectoring	120-degree sectoring
Cell splitting	Selective cell splitting

3. JUSTIFICATION FOR CLUSTER SIZE

The cluster size is calculated using:

$N = i^2 + ij + j^2$

Taking i = 2 and j = 2:

$N = 2^2 + (2 \times 2) + 2^2$

$N = 4 + 4 + 4$

$N = 12$

The number of channels per cell is:

Channels per cell = Total channels / Cluster size

$C = 336 / 12$

$C = 28$ channels

Cluster size 12 provides fewer channels per cell than smaller clusters but improves frequency reuse distance and reduces co-channel interference.

4. FREQUENCY REUSE DISTANCE

The frequency reuse distance is:

$D = R \times \sqrt{(3N)}$

$D = 0.187 \times \sqrt{(3 \times 12)}$

$D = 0.187 \times 6$

$D = 1.122$ km (approximately)

Therefore, the frequency reuse distance is approximately 1.122 km.

5. SIGNAL-TO-INTERFERENCE RATIO

The simplified SIR formula for six first-tier interferers is:

SIR = (3N)^(n/2) / 6

For N = 12 and n = 2.5:

SIR = 36^1.25 / 6

SIR ≈ 14.70

SIR (dB) = 10 × log10(14.70)

SIR ≈ 11.67 dB

This is slightly below the selected 12 dB target.

With idealized 120-degree sectoring, assuming two dominant interferers:

SIR = 36^1.25 / 2

SIR ≈ 44.10

SIR (dB) = 10 × log10(44.10)

SIR ≈ 16.44 dB

Therefore, sectoring helps the recommended design meet the 12 dB target under the simplified open-campus model.

6. NUMBER OF CELLS REQUIRED

For 100,000 subscribers:

Traffic demand = Number of subscribers × Traffic per subscriber

Traffic demand = 100,000 × 0.025

Traffic demand = 2,500 Erlangs

For 28 channels per cell, the Erlang-B model gives approximately 20.15 Erlangs per cell at 2% blocking.

Required cells = 2,500 / 20.15

Required cells ≈ 124.07

Rounding up to a complete cluster of 12 cells:

Required cells = 132 cells

The approximate cell radius is:

R = √[Total area / (2.598 × Number of cells)]

R = √[12 / (2.598 × 132)]

R ≈ 0.187 km

Therefore, approximately 132 cells are required under these idealized traffic and coverage assumptions.

7. ESTIMATED NETWORK CAPACITY

Total capacity = Number of cells × Traffic capacity per cell

Total capacity = 132 × 20.15

Total capacity ≈ 2,659.8 Erlangs

Number of supported subscribers:

Subscribers = Total capacity / Traffic per subscriber

Subscribers = 2,659.8 / 0.025

Subscribers ≈ 106,392

Therefore, the estimated network capacity is sufficient for approximately 100,000 subscribers under the assumed model.

8. RECOMMENDED NETWORK STRATEGIES

A. FREQUENCY REUSE

Use cluster size 12 to provide a balance between channel allocation and interference management.

B. SECTORING

Use directional antennas to reduce co-channel interference and improve signal quality.

C. DYNAMIC CHANNEL ASSIGNMENT

Allocate available channels according to current traffic demand to reduce congestion.

D. SELECTIVE CELL SPLITTING

Divide only high-traffic cells into smaller cells to increase capacity without unnecessary infrastructure expansion.

E. MACRO–MICRO CELL HIERARCHY

Use macro cells for wider coverage and micro cells in crowded locations such as classrooms, libraries, canteens and stadiums.

9. COST AND PERFORMANCE TRADE-OFFS

Design decision	Benefit	Limitation
Cluster size 12	Better frequency separation	Fewer channels per cell
Sectoring	Reduces interference	Requires directional antennas
Smaller cells	Increases capacity	More base stations
Dynamic allocation	Better channel utilisation	More complex management
Selective splitting	Reduces unnecessary cost	Requires traffic monitoring

The network requires additional base stations and maintenance as the subscriber population grows. Selective cell splitting helps control these costs.

10. FUTURE SCALABILITY

The recommended network can support increasing subscriber demand through:

1. Adding microcells in high-traffic locations.
2. Using dynamic channel allocation.
3. Monitoring traffic during busy hours.
4. Optimising antenna direction and transmission power.
5. Performing selective cell splitting when congestion increases.

These methods improve capacity without requiring additional spectrum.

11. CONCLUSION

The recommended cellular network uses a cluster size of 12, 28 channels per cell, 120-degree sectoring and selective cell splitting. The estimated design requires approximately 132 cells to support 100,000 subscribers at a maximum 2% blocking probability under the stated assumptions. Sectoring improves the theoretical SIR to approximately 16.44 dB in the open-campus model, exceeding the selected 12 dB target. The combination of frequency reuse, dynamic channel assignment and macro–micro cell hierarchy provides a scalable and efficient solution for the university township.

Note: These are planning estimates based on idealized hexagonal coverage, Erlang-B traffic and simplified interference assumptions. Actual deployment requires detailed radio propagation, handoff, antenna and site-planning simulations.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8

Marks Changed Requirement

A convocation / sports event will bring 25,000 attendees, each generating 0.05 Erlang, into a 1 km² stadium zone for four hours. Outside the event, demand stays as originally forecast. Event days occur roughly twice a month.

Re-evaluate your original design under the changed conditions.
You may:
RETAIN → MODIFY→ or REDESIGN

your original solution.

A. Impact Analysis – 2

Marks Guiding Enquiry

What exactly has changed, and which of your original assumptions no longer hold?

- Estimate the new offered traffic density and compare it with the design density.
- Identify affected assumptions (uniform traffic, steady demand, permanent capacity).
- analyse why the short duration and predictability of the surge matter for the choice of solution.

Evidence Expected

A clear analysis of how the surge affects the original design.

B. Evaluation of the Original Design – 2

Marks Guiding Enquiry

How would your recommended design behave on event day?

- Estimate blocking probability and interference problems in the stadium zone using your own model.
- Assess whether the design can be reused unchanged and what fails first.
- Identify whether an alternative strategy handles the surge more economically.

Evidence Expected

A technically supported evaluation of the original design under the surge.

C. Adaptation or Redesign – 2

Marks Guiding Enquiry

Should the township network be altered permanently, temporarily or not at all?

- Choose retain, modify or redesign (e.g., micro-cells/cell splitting in the stadium zone, 60° sectoring, temporary cells-on-wheels, dynamic channel borrowing).
- Describe the revised strategy and what changes in reuse pattern, power and handoffs.
- Determine the effect on capacity, SIR and planning cost.

Evidence Expected

A design response with clear radio-planning reasoning and quantified effect.

D. Evidence-Based Defence – 2

Marks Guiding Enquiry

Why is your final decision the right one, given cost, frequency of events and the evidence you generated?

- Use analytical results, simulation results, surge size and duration, planning cost and trade-offs.
- Show that the decision follows logically from your evidence.

Evidence Expected

A well-supported final decision integrating theory, simulation and the changed constraints.

ANSWER

TASK 12 – DESIGN A CELLULAR NETWORK FOR A LARGE-SCALE EVENT IN THE UNIVERSITY TOWNSHIP

1. INTRODUCTION

A large-scale event creates a sudden increase in mobile network traffic within a limited geographical area. During university events, festivals, sports competitions and public gatherings, thousands of users may access the network simultaneously.

The university township has to support an event with 25,000 attendees in a 1 km^2 stadium zone for four hours. The event takes place approximately twice every month.

The objective is to provide sufficient network capacity, reduce call blocking and maintain good signal quality without increasing the available spectrum.

2. EVENT REQUIREMENTS

The given event requirements are:

Parameter	Value
Number of attendees	25,000
Stadium area	1 km^2
Busy-hour traffic per attendee	0.05 Erlang
Event duration	4 hours
Event frequency	Twice monthly
Available duplex channels	336
Maximum blocking probability	2%

3. CALCULATION OF EVENT TRAFFIC DEMAND

Total traffic demand is calculated using:

Total traffic = Number of attendees × Traffic per attendee

Total traffic = 25,000 × 0.05

Total traffic = 1,250 Erlangs

Therefore, the stadium requires 1,250 Erlangs of traffic capacity during the event.

4. EVALUATION OF THE EXISTING NETWORK

The existing network was designed for normal university traffic.

During the event:

- 25,000 additional users access the network.
- Traffic demand increases significantly.
- Available channels become occupied quickly.
- Interference may increase because of concentrated users.
- Call blocking and dropped calls may occur.
- Data and voice services may become congested.

The existing network is not sufficient to support the sudden event demand without additional capacity planning.

5. CALCULATION OF REQUIRED CHANNELS

The Erlang-B formula is used to calculate the number of channels required to support 1,250 Erlangs at a 2% blocking probability.

Erlang-B formula:

$B(A,C) = (A^C / C!) / [\Sigma (A^k / k!), \text{from } k = 0 \text{ to } C]$

Where:

- A = Offered traffic in Erlangs
- C = Number of channels
- B = Blocking probability

Given:

A = 1,250 Erlangs

B = 0.02

Using the Erlang-B model, approximately 1,306 traffic channels are required.

However, only 336 duplex channels are available.

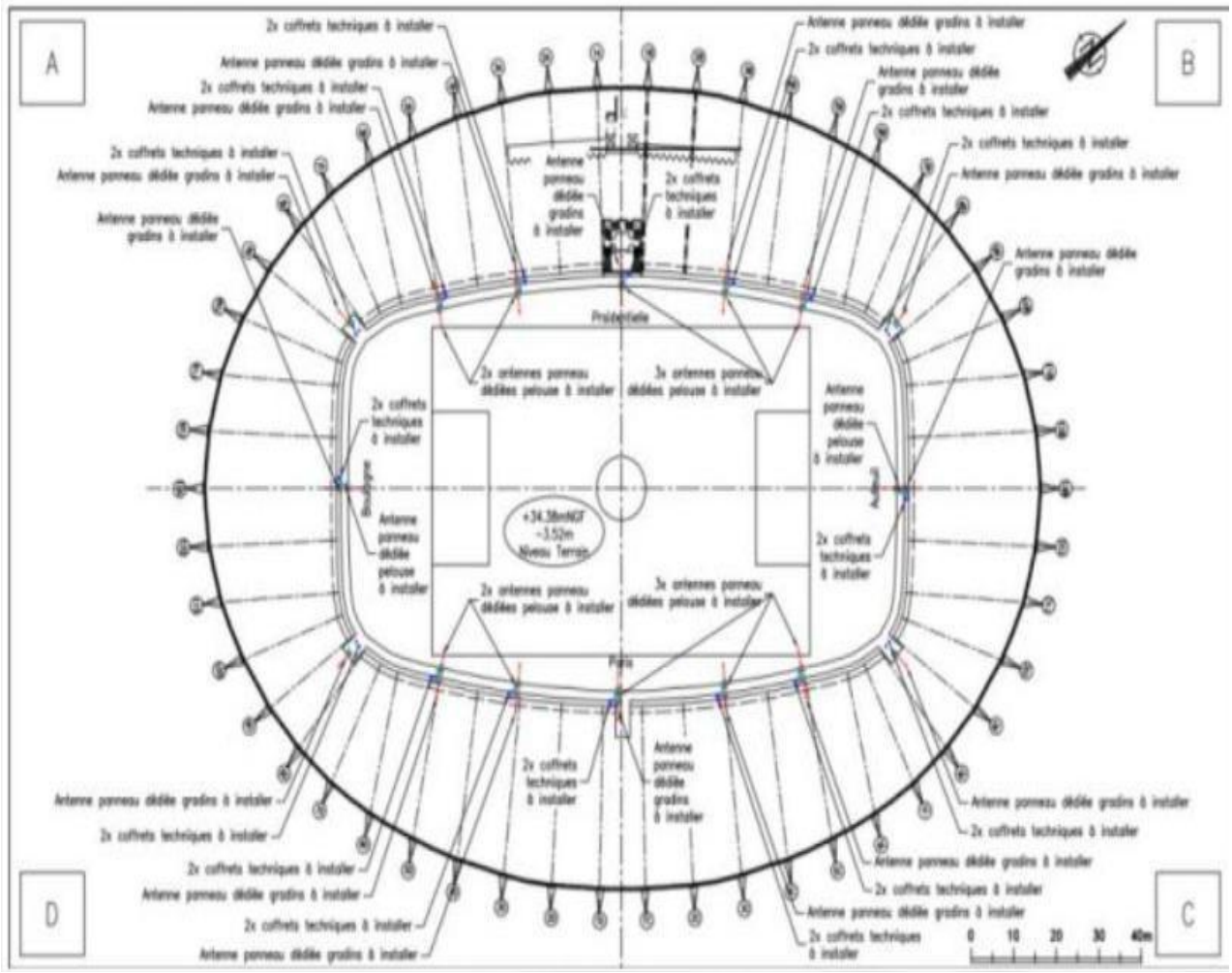
Therefore, the network cannot support all event traffic using a single conventional channel allocation.

6. PROPOSED EVENT NETWORK DESIGN

A temporary high-capacity network should be deployed in the stadium area.

The proposed design includes:

1. Temporary microcells around the stadium.
2. Sectorised antennas to reduce interference.
3. Dynamic channel assignment to distribute traffic.
4. Small cells near crowded seating areas.
5. Power control to reduce unnecessary interference.
6. Temporary base stations for additional spatial reuse.



7. PROPOSED CELL SPLITTING AND COVERAGE

The stadium area is 1 km².

Assume a temporary microcell radius of 0.10 km.

Hexagonal cell area:

$Ac = 2.598 \times R^2$

$Ac = 2.598 \times 0.10^2$

$Ac = 0.02598\ km^2$

Number of cells:

M = Total area / Area of one cell

M = 1 / 0.02598

M = 38.49

Rounding up:

M = 39 cells

Therefore, approximately 39 microcells are required for idealized coverage of the stadium.

8. FREQUENCY AND CHANNEL ALLOCATION

The available 336 duplex channels must be reused efficiently.

For a cluster size of 7:

Channels per cell = 336 / 7

Channels per cell = 48

The 39 microcells can use repeated frequency reuse patterns with proper spatial separation and interference control.

Sectoring can further improve interference performance by using directional antennas.

Dynamic channel assignment can allocate available channels according to traffic demand.

9. CAPACITY IMPROVEMENT METHODS

A. TEMPORARY MICROCELLS

Install additional microcells in crowded stadium locations to increase spatial frequency reuse.

B. SECTORING

Divide coverage into directional sectors to reduce co-channel interference.

C. DYNAMIC CHANNEL ASSIGNMENT

Allocate channels based on real-time traffic demand.

D. LOAD BALANCING

Distribute users across nearby base stations to avoid congestion in a single cell.

E. POWER CONTROL

Adjust transmission power to reduce unnecessary interference between neighbouring cells.

10. FOUR-HOUR EVENT MANAGEMENT PLAN

Event period	Network management activity
Before event	Install and test temporary microcells
First hour	Monitor user connections and traffic
Second hour	Adjust channel allocation and power
Third hour	Monitor congestion and balance network load
Fourth hour	Maintain service and prepare for user departure
After event	Remove temporary equipment and analyse performance

11. COST AND PRACTICAL CONSIDERATIONS

The temporary network requires:

- Temporary base stations.
    - Directional antennas.
    - Backhaul connectivity.
    - Power supply.
    - Network monitoring equipment.

- Technical support staff.

Since the event occurs twice monthly, temporary or deployable network equipment can be more economical than permanently expanding the entire township network.

12. FINAL RECOMMENDATION

The existing network should be modified temporarily during the event rather than relying only on its normal configuration.

The recommended solution is:

- Deploy approximately 39 microcells for idealized stadium coverage.
- Use cluster size 7 as an initial frequency reuse planning assumption.
- Apply sectoring to reduce co-channel interference.
- Use dynamic channel allocation and load balancing.
- Monitor traffic continuously during the four-hour event.
- Provide additional temporary radio capacity through spatial reuse.

IMPORTANT: The 39-cell estimate provides a coverage starting point, not proof of sufficient capacity. With 1,250 Erlangs of demand, the network needs additional radio resources and carefully planned frequency reuse. The 336-channel limit cannot be overcome simply by installing more cells; interference and channel reuse must also be controlled.

13. CONCLUSION

The large-scale university event creates a traffic demand of 1,250 Erlangs from 25,000 attendees. The existing network may experience congestion and increased call blocking during this period. A temporary network using microcells, sectoring, dynamic channel assignment and load balancing can improve coverage and capacity. Approximately 39 microcells are estimated for stadium coverage using a 0.10 km radius. Proper frequency reuse and interference management are essential to provide reliable communication throughout the four-hour event without increasing the available spectrum.

MARK DISTRIBUTION

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TOTAL	**50**